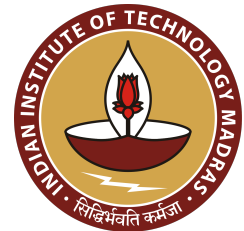


BLAST: Heuristic Alignment Algorithms

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- It is a method to compare two sequences

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 - Global Alignment - Comparing two sequences in their entirety
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- There can be multiple local alignments between sequences

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Question

For comparing sequences which alignment do we use - local or global?

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Answer

Local!

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Answer

Local!

- Why? Because when comparing distant sequences, it is more likely that a given sequences will be conserved at small regions rather than the entire length of the sequences

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- The time complexity of one Smith-Waterman alignment run is $\mathcal{O}(qd)$
- For the entire database: $\mathcal{O}(nqd)$
- The number of sequences in a database (n) is extremely large, therefore running multiple alignments is time consuming.
- **Solution:** Use Heuristics

Definition: Heuristics

Heuristic are strategies designed to efficiently tackle complex problems by providing approximate solutions when exact methods are impractical.

Basic Local Alignment Search Tool (BLAST)

- The Basic Local Alignment Search Tool (BLAST) was first described in 1990 by Altschul et al. to allow fast database searches
- The goal: Given a query sequence, Q , of length q , and a database sequence, D , of length n , find all **ungapped local alignments** with score at least S_{cutoff} .
- Here, S is the alignment score, calculated using a scoring matrix



Example scoring matrix for DNA comparison

	A	T	G	C
A	+2	-1	+1	-1
T	-1	+2	-1	1
G	1	-1	+2	-1
C	-1	+1	-1	+2

Maximal Segment Pair

- Before proceeding to understand BLAST, a few definitions:

Definition: Maximal Segment Pair (MSP)

A maximal segment pair (MSP) is the highest-scoring **ungapped local alignment** between **two equal-length substrings**. An MSP is **locally maximal** if the score cannot be increased by extending or shortening the alignment.



Example

INFORMATION
AUTOMATION

- The goal of BLAST is to identify all **Locally Maximum Segment Pairs** between the query and database sequence. These are also known as **High Scoring Pairs (HSPs)**

Definition: k -mer and k -mers

k -mer is a contiguous string of length k . k -mers is a set of all the contiguous substrings of length k that are contained in a given string

- k -mers for a sequence “TATGGGGTGC”
- $k=1$, {T, A, T, G, G, G, G, T, G, C}
- $k=2$, {TA, AT, TG, GG, GG, GG, GT, TG, GC}
- $k=4$, {TATG, ATGG, TGGG, GGGG, GGGT, GGTG, GTGC}
- What is the number of possible k -mers in a string of length n ?

Inputs : Q - query of length q , D - database sequence of length d

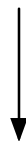
Parameters :

- Scoring matrix,
- S Minimum MSP score,
- k - length of k -mer,
- T - minimum score for valid k -mer,

Output : All MSPs with score $\geq S_{\text{cutoff}}$

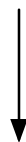
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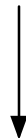


Seeding: Identifying all exact matches of L in D . This is the heuristic.

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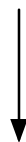


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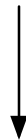


Extension: Extend alignments to find Local MSPs with score at least S_{cutoff}

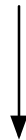
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Extension: Extend alignments to find Local MSPs with score at least S_{cutoff}



Stopping: Stop when score drops or query ends

Example 1 - DNA sequence

- Let's take a simple example first:

Inputs:

- Query sequence: ATGCAT (length $q = 6$)
- Database sequence: CATGCATG (length $d = 8$)

Parameters:

- Scoring: match = +2, mismatch = -2
- k -mer length, $k = 3$
- Minimum k -mer score, $T = 5$
- Minimum MSP score, $S_{\text{cutoff}} = 10$

Example 1 - Preprocessing

- The input query Q : ATGCAT
- All the k -mers in the query:

Position	Query k -mer	Score upon match	Keep? (Score $\geq T(5)$)
Pos 1–3	ATG	+2+2+2=6	Yes
Pos 2–4	TGC	+2+2+2=6	Yes
Pos 3–5	GCA	+2+2+2=6	Yes
Pos 4–6	CAT	+2+2+2=6	Yes

- For simplicity, in this example, assume that we only have **perfect matches**

Example 1 - Seeding

- Matching all k -mers to the database sequence

CATGCATG
ATG

CATGCATG
TGC

CATGCATG
GCA

CATGCATG
CAT

Example 1 - Seeding and Extension

- Extend on both sides till score decreases or you reach query end

CATGCATG
TGC

$$S_{\text{current}} = 6$$

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CATGCATG
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Extend right

CATGCATG
TGCA

$$S_{\text{current}} = 8$$

Example 1 - Seeding and Extension

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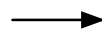
CATGCATG
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Extend right

CATGCATG
TGCA

$$S_{\text{current}} = 8$$



CATGCATG
TGCAT

$$S_{\text{current}} = 10$$

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CATGCATG
TGC

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CATGCATG
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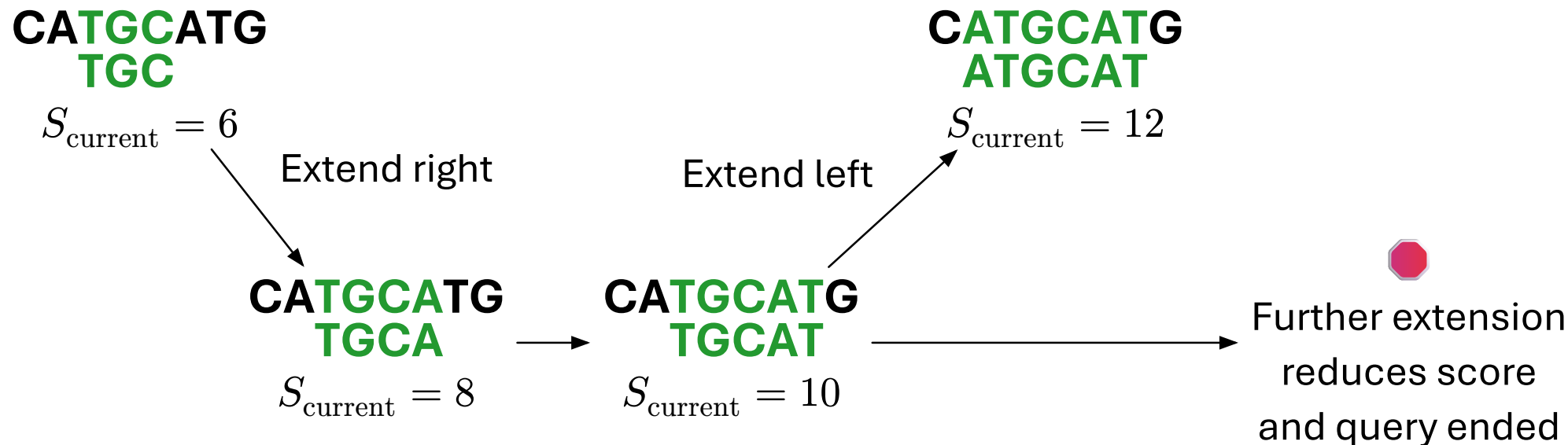
CATGCATG
TGCAT

$$S_{\text{current}} = 10$$


Further extension
reduces score
and query ended

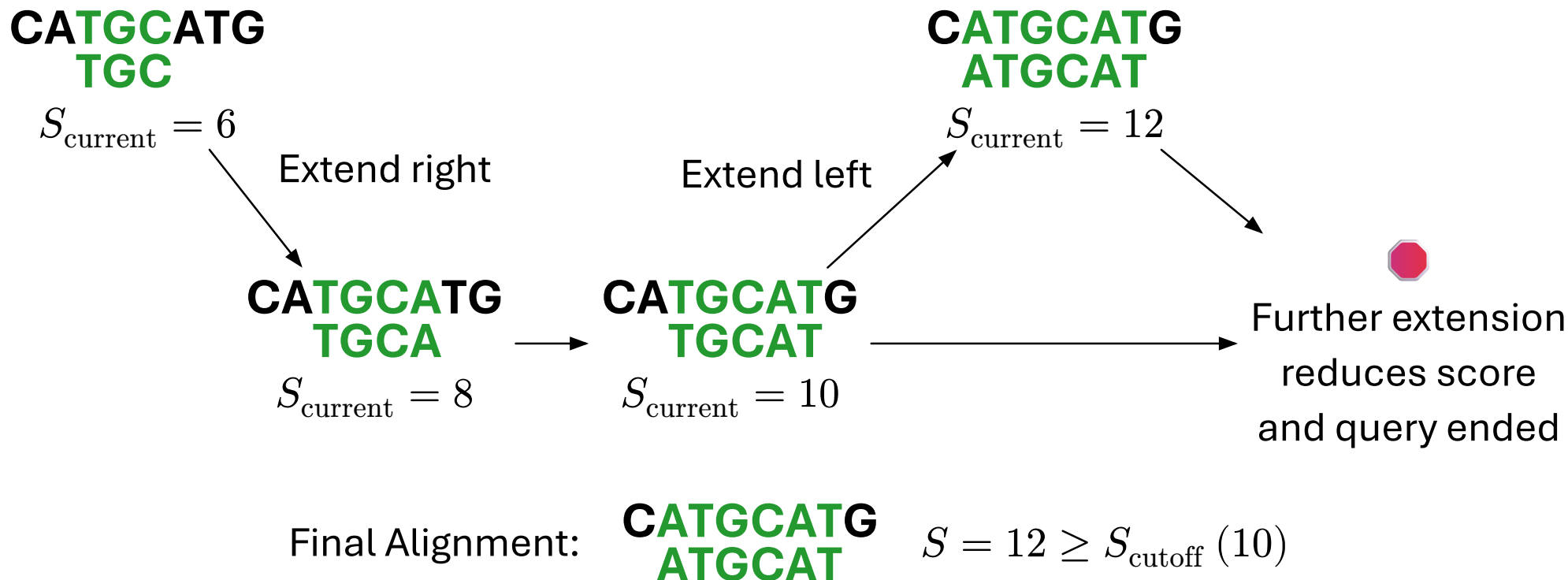
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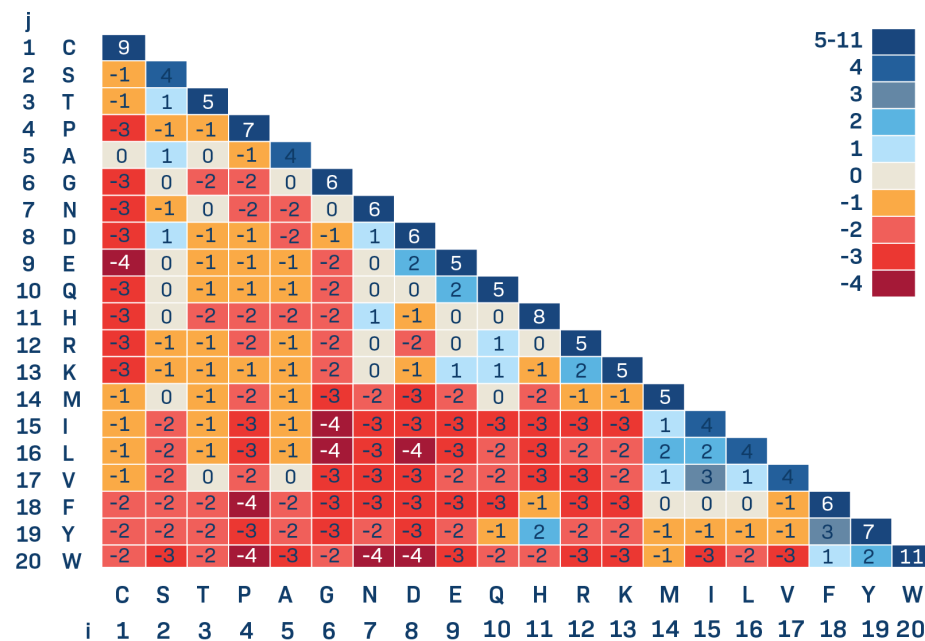
Example 2 - Protein Sequence, slightly more complicated

Inputs:

- Query sequence: MALWDY (length $q = 6$)
- Database sequence: PMALYYG (length $d = 7$)

Parameters:

- Scoring: BLOSUM62
- k -mer length, $k = 3$
- Minimum k -mer score, $T = 10$
- Minimum MSP score, $S_{\text{cutoff}} = 13$



BLOSUM62 Scoring matrix
(Source: LabXchange.com)

Example 2 - Preprocessing

- The input query Q : MALWDY
- Let's focus on one k -mer: ALW
- Now identify the k -mers of the same length that align and have a score greater than $T = 10$. Some examples:

Query k -mer	Target k -mer	Score	Keep? (Score $\geq T(10)$)
ALW	ALW	+4+4+11=19	Yes
ALW	ALF	+4+4+1=9	No
ALW	SLW	+1+4+11=16	Yes
ALW	AMW	+4+2+11=17	Yes
ALW	ALY	+4+4+2=10	Yes

Example 2 – Protein Seeding and Extension (BLASTP)

- Extend using substitution matrix scores (e.g., BLOSUM62) until score decreases or sequence end

PMALY YG
ALW

$$S_{\text{current}} = 10$$

BLOSUM62: A-A(+4), L-L(+4), W-Y(+2)

Example 2 – Protein Seeding and Extension (BLASTP)

- Extend using substitution matrix scores (e.g., BLOSUM62) until score decreases or sequence end

PMALY YG
ALW

$$S_{\text{current}} = 10$$

BLOSUM62: A-A(+4), L-L(+4), W-Y(+2)

Extend left

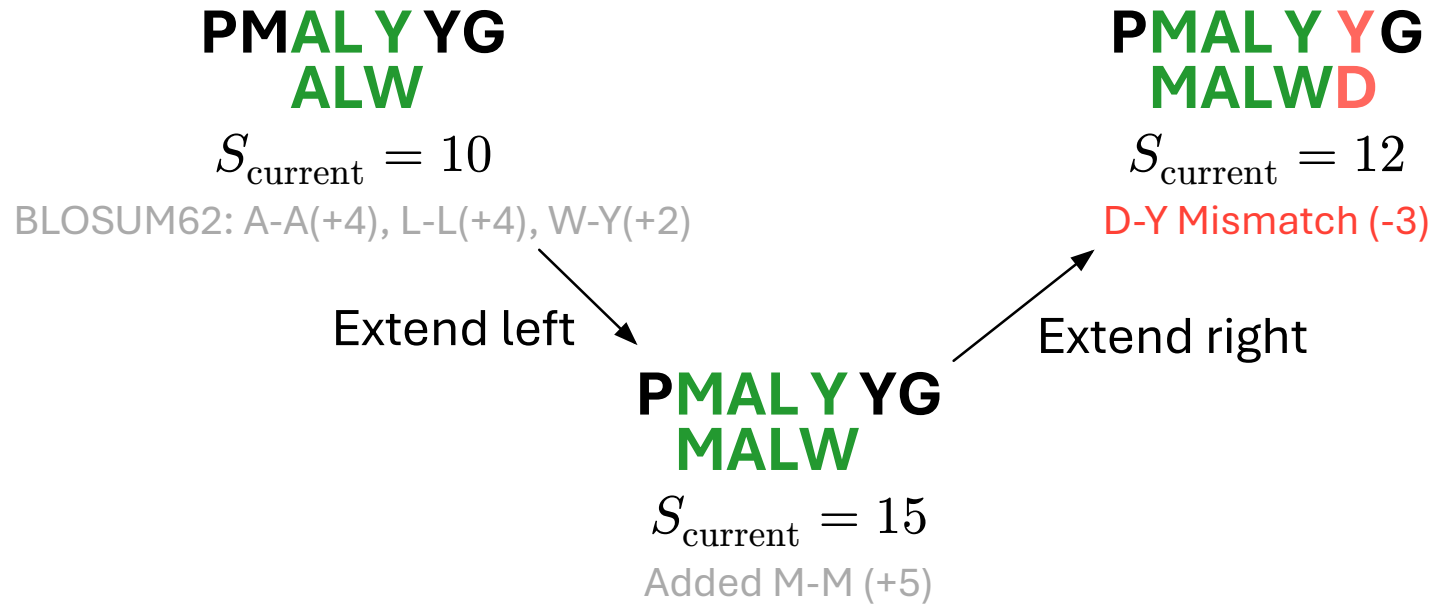
PMALY YG
MALW

$$S_{\text{current}} = 15$$

Added M-M (+5)

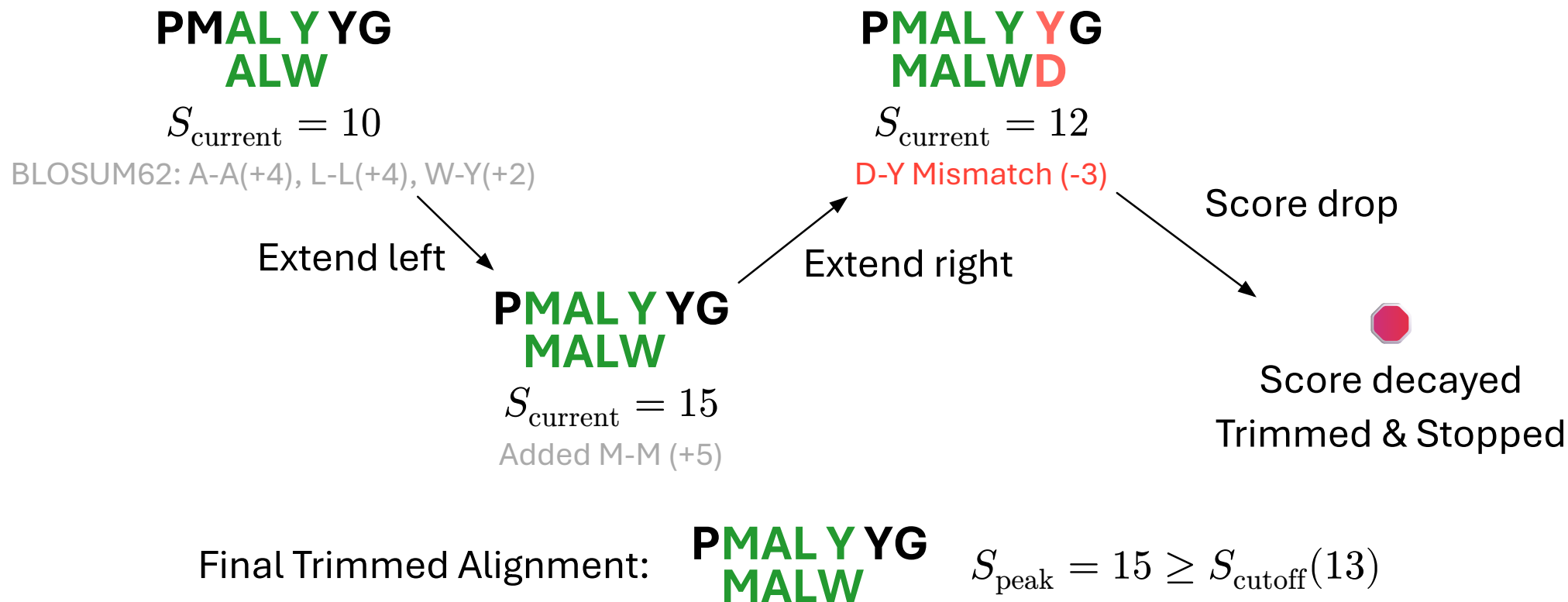
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Next steps: How significant is an alignment?

- How does one differentiate from a genuine alignment versus something that could have come up by chance?
- We need to verify that the chance of our result is greater than the chance if the query and database sequences were random

What is the probability of a single alignment by chance?

- Consider an alignment of two random sequences with a score S , what is the probability of $S > S_{\text{cutoff}}$
- Intuitively, as S_{cutoff} increases, the chance of random sequence alignment producing this score drastically reduces
- We can model this using an exponential distribution

Exponential Distribution

$$f(x; \lambda) = \lambda e^{-\lambda x}; \quad \lambda > 0, x \in [0, \infty)$$

Now, the probability of obtaining a score of at least S_{cutoff} can be calculated using the survival function:

$$P(S \geq S_{\text{cutoff}}) = 1 - \text{CDF}(x) = e^{-\lambda S_{\text{cutoff}}}$$

How do we identify the correct λ

- Recall the probability of obtaining score of at least S is $P(S \geq S) = e^{-\lambda S}$
- S is score that is decided using a scoring matrix.

How do we identify the correct λ

- Recall the probability of obtaining score of at least S is $P(S \geq S) = e^{-\lambda S}$
- S is score that is decided using a scoring matrix. But scoring matrices are **arbitrary**

Consider the two cases:

 Case 1	 Case 2
Scoring: Match: +2; mismatch and gaps: -1	Scoring: Match: +4; mismatch: -1 and gaps: 0
ATG ACC	ATG - - A- - CC
Final Score is 2	Final Score is 2

- Which is the correct alignment?

A principled way to think about alignment scoring

- A measure of sequence similarity is a quantification of how likely two residues (DNA/RNA/Protein) align with one another

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- A measure of sequence similarity is a quantification of how likely two residues (DNA/RNA/Protein) align with one another
 - Therefore we can take a probabilistic approach to quantifying sequence similarity
- Probabilities of alignment between two residues can be quantified by studying previously aligned sequences

A concrete example

If one studies 100,000 aligned columns in evolutionarily related proteins, and Leucine (L) is aligned with Isoleucine (I) 3000 times, then

- the probability of leucine and isoleucine aligning, $p_{(L, I)} = 0.03$
- the marginal probabilities of leucine and isoleucine in the data are p_L and p_I respectively
- Odds of seeing aligned Leucine and Isoleucine aligned in a given sequence:

$$\text{Odds} = \frac{p_{(L, I)}}{p_L p_I}$$

- $\text{Odds} > 1 \Rightarrow$ the alignment is less likely to have occurred due to random chance
- $\text{Odds} \leq 1 \Rightarrow$ the alignment is more likely to have occurred due to random chance
- As a convention, we report the logarithm odds

Probabilistic Interpretation

- Generalizing, the log-odds “score” of similarity between two residues (i, j) in an alignment is:

$$s_{ij} = \ln \left(\frac{q_{ij}}{p_i p_j} \right)$$

- Rearranging, we get: $q_{ij} = p_i p_j e^{s_{ij}}$. And since, q_{ij} is a probability $\sum_{ij} q_{ij} = 1$.
- Now, let's consider an arbitrary scoring paradigm s'_{ij} and the relation between s_{ij} and s'_{ij} is $s_{ij} = \lambda s'_{ij}$, then

$$q = \sum_{ij} p_i p_j e^{\lambda s'_{ij}} = 1$$

- λ , therefore is the parameter that helps us convert an arbitrary score to an universal score based on probability
- In practice, λ is pre-calculated for different scoring matrices

What is the number of alignments expected between random sequences?

Let y be defined as the Expected number of alignments for given query sequence Q (of length q) and database sequence D (of length d) and cutoff score S_{cutoff} .

- Intuitively:
 - $y \propto qd$, because the longer sequences, more likely that residues match randomly
 - $y \propto \frac{1}{S}$, because the higher the score, the less likely that alignment random residues reach that value
 - Since we consider those alignments that satisfy $S \geq S_{\text{cutoff}}$, $y \propto P(S \geq S_{\text{cutoff}})$

Bringing it all together:

$$y = K q d e^{-\lambda S_{\text{cutoff}}}$$

where K is a scaling parameter. A higher y means that one expects more random matches.

How statistically significant are the alignments?

The value y is the average number (or rate) of hit for the given sequences and whose alignment has a score S is:

$$y = Kqde^{-\lambda S}$$

For calculating the the probability for c matches, model using the Poisson distribution.

Possion Distribution

$$P(x = k; \lambda) = \frac{\lambda^k e^{-\lambda}}{k!}$$

- The Poisson distribution models the number of discrete, independent random events that occur during a fixed interval of time or space at a constant average rate

How statistically significant are the alignments?

- In our case, the probability of finding a random alignment in the **space** of a sequence alignment follows a Poisson distribution (Karlin & Altschul, 1990)
- For given query sequence Q (of length q) and database sequence D (of length d) and cutoff score S_{cutoff} , how do we calculate if the obtained alignments are non-random?

To calculate significance - calculate probability of at least one random alignment

$$\begin{aligned} P(x \geq 1) &= 1 - P(x = 0) \\ &= 1 - \frac{y^0 e^{-y}}{0!} \\ &= 1 - e^{-y} \end{aligned}$$

Let's consider an example

Consider the following example:

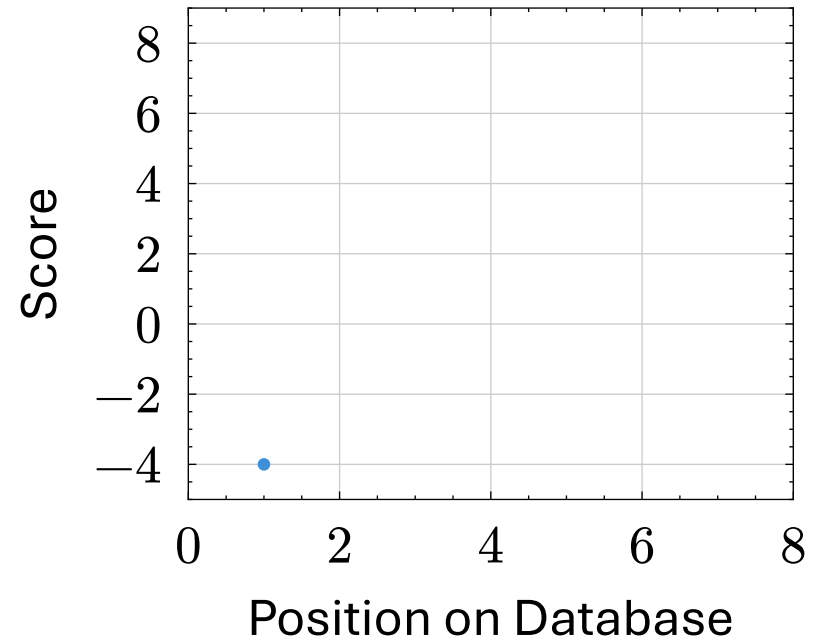
- Query: ATAT
- Database sequence: CATATATGGG
- Scoring: match - +2, mismatch - -1
- What happens as we move the alignment of query sequence along the database?

Let's consider an example

C A T A T A T G G G
A T A T

$S = -4$

Scored →

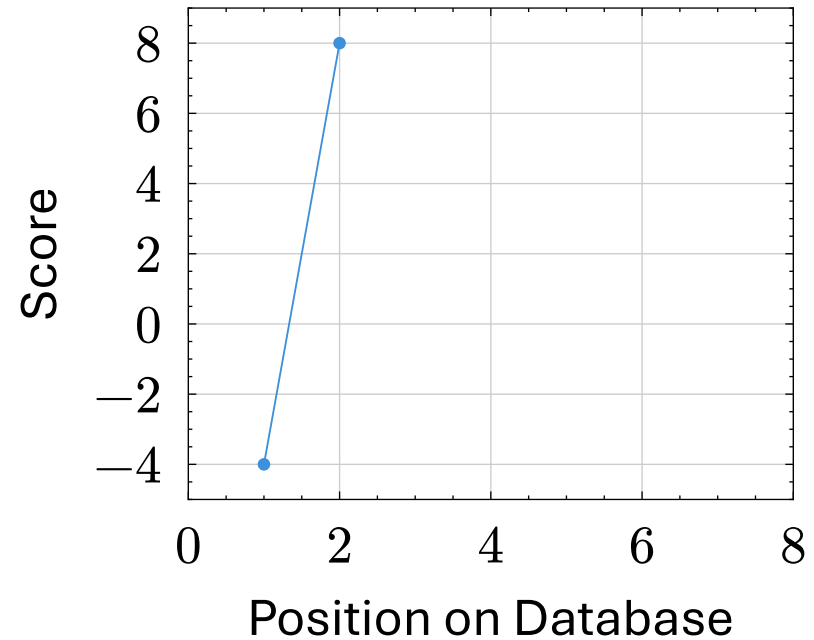


Let's consider an example

C A T A T A T G G G
A T A T

$S = 8$

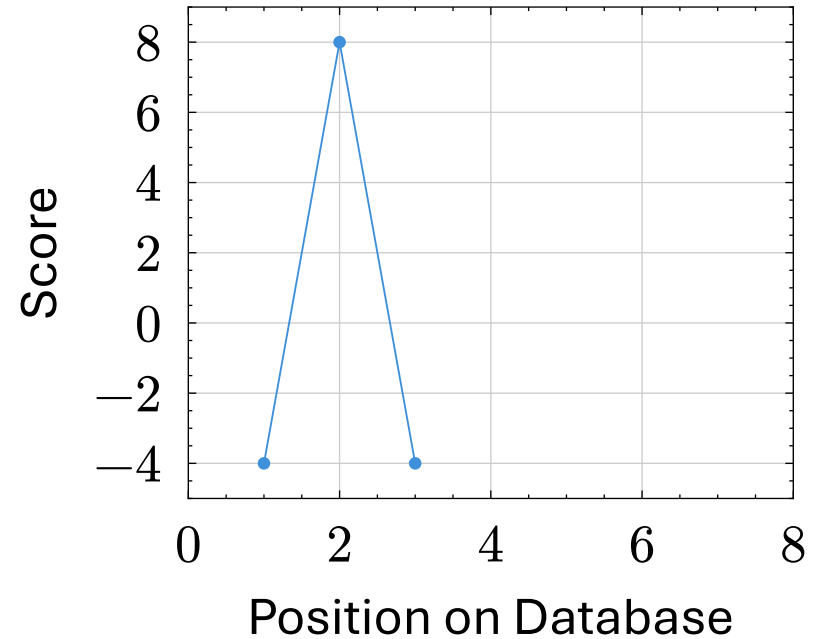
Scored →



Let's consider an example

C A T A T A T G G G
 A T A T
 $S = -4$

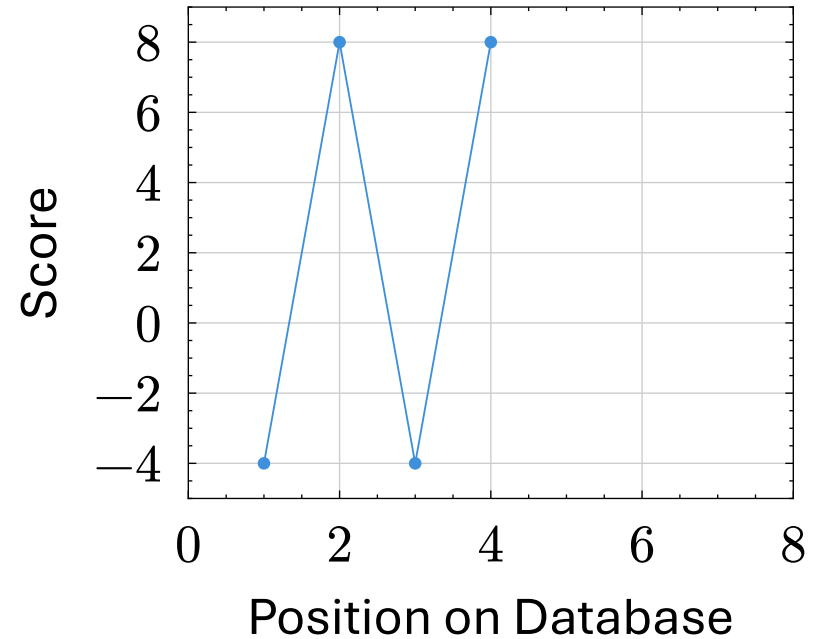
Scored →



Let's consider an example

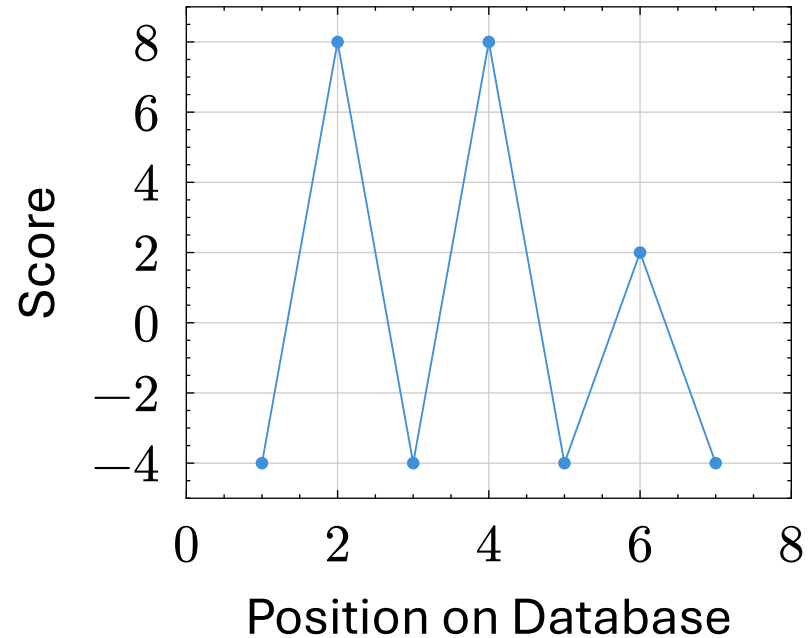
C A T A T A T G G G
 A T A T
 A T A T
 $S = 8$

Scored →



Let's consider an example

C A T A T A T G G G
 A T A T Scored
 $S = -4$



? Question

Are there really 2 high scoring alignments?

Overlapping alignments

- Consider the two high scoring alignments of the examples again:

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- The center AT is repeated in both the cases.
- This means that they cannot be treated as separate alignments
- And this is exactly what K will correct

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- Recall the expected number of random alignments: $y = Kqde^{-\lambda S_{\text{cutoff}}}$

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- K fixes the **search-space axis** — it converts the nominal search space qd into an *effective* one

What is K actually doing?

- Recall the expected number of random alignments: $y = Kqde^{-\lambda S_{\text{cutoff}}}$
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- K fixes the **search-space axis** — it converts the nominal search space qd into an *effective* one

Definition: The parameter K

K is the scaling parameter that converts the nominal search space qd into the **effective number of independent positions** at which a locally maximal alignment could begin. Typically $K < 1$.

How do we determine K?

K can be calculated empirically (Altschul & Gish, 1996):

Steps:

- Randomly generate pairs of queries and database sequences
- Use Smith-Waterman algorithm to identify the local alignment between the two sequences and store the maximum alignment scores $S_{\max}$
- Altschul & Gish (1996), showed that the maximum score follows an extreme value gumbel distribution

Extreme Value Gumbel Distribution

$$F(x) = \exp\left(-\exp\left(-\frac{x - \mu}{\beta}\right)\right)$$

How do we determine K?

Extreme Value Gumbel Distribution

$$F(x) = \exp\left(-\exp\left(-\frac{x - \mu}{\beta}\right)\right)$$

- In the case of local alignment, $\mu = \frac{\ln(Kmn)}{\beta}$

Which means,

$$F(x) = P(S_{\max} \leq x) = \exp(-K m n e^{-\lambda x})$$

Solving the above equation will give us the required value of K .

BLAST calculates a universal score, $S' = \frac{\lambda S - \ln K}{\ln 2}$, which is then reported instead of unnormalized scores.

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- Real homologies contain insertions and deletions. An ungapped extension cannot capture that effectively.

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 - Combine their significance
- The problem with that:
 - The combined result is only significant if you find **every** constituent HSP
 - Miss one and the whole hit is lost
 - To avoid missing them you must keep T low — many hits, many extensions, slow

Example 3 - When ungapped alignment fails

Inputs:

- Query sequence: ACGTTGCA (length $q = 8$)
- Database sequence: ACGTAATGCA (length $d = 10$)

Parameters:

- Scoring: match = +2, mismatch = -2
- Gap cost (simplified affine): $-(3 + 1 \times \text{length})$ (Original BLAST does not have gaps)
- k -mer length, $k = 3$; minimum k -mer score, $T = 6$
- Minimum MSP score, $S_{\text{cutoff}} = 10$

Example 3 - Ungapped BLAST finds two HSPs

ACGTAAATGCA
ACGT
Seed CGT $S = 8$

Example 3 - Ungapped BLAST finds two HSPs

ACGTAATGCA
ACGT

Seed CGT $S = 8$

ACGTAATGCA
TGCA

Seed TGC $S = 8$

- Neither reaches $S_{\text{cutoff}} = 10$. **Ungapped BLAST reports nothing.**

Example 3 - A gap rescues the alignment

- Gapped BLAST joins the two HSPs across the diagonal shift:

ACGT**A**TGCA
ACGT- -TGCA

$$S = 8 \times (+2) - (3 + 2 \times 1) = 16 - 5 = 11 \geq S_{\text{cutoff}}(10)$$

- The **same evidence**, assembled differently, crosses the threshold
- The joined score is 11, not 16 — the two-residue gap is charged -5
- But how do we include gaps in BLAST?

Paying for it: the two-hit method

- A gapped DP extension costs far more than an ungapped one. Two moves recover the time.

Definition: The two-hit rule

Invoke an ungapped extension only when **two non-overlapping word hits of score $\geq T$ lie on the same diagonal, within distance A of one another**. The extension starts from the second hit.

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- Defaults for Protein BLAST: $k = 3$, $T = 11$, $A = 40$

Does the two-hit method work?

- The paper's worked case: broad bean leghemoglobin I vs horse β -globin (Altschul, 1997)

Heuristic	T	Word hits found	Extensions triggered
One-hit	13	15	15
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Why this works

A genuine HSP long enough to matter almost always contains **two** word hits. An isolated hit almost never grows into anything. Requiring two is a cheap test that discards nearly all the noise before paying for an extension.

Extending Examples 1 and 2: the X -drop rule

- Our earlier examples said: *extend until the score decreases*
- **BLAST does not do this.** A single mismatch would terminate almost every real alignment.

Definition: X -drop extension

Keep extending while

$$S_{\text{current}} > S_{\text{best so far}} - X$$

then stop and **trim back to the best-scoring position.**

X is a tolerance for temporary dips in score.

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Answer

Smith-Waterman Algorithm

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Full steps:

- **Seed:** Match two kmers of the query with the database within A distance of one another

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Answer

Smith-Waterman Algorithm

Full steps:

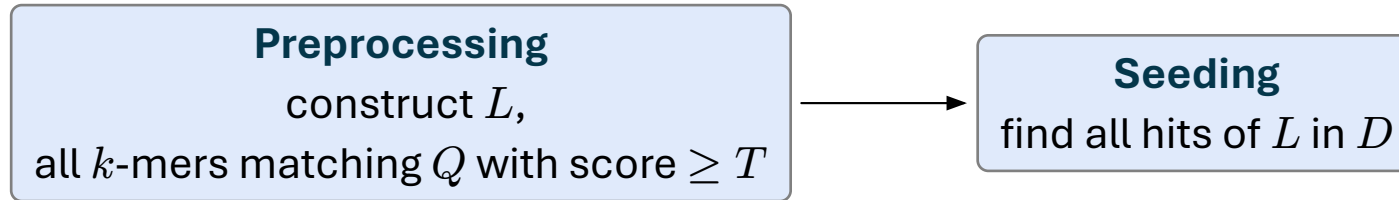
- **Seed:** Match two kmers of the query with the database within A distance of one another
- **Extend:** run Smith-Waterman algorithm outward in both directions from the two sides
- **Result:** Alignment flanked by the two seed sequences

The full gapped BLAST pipeline

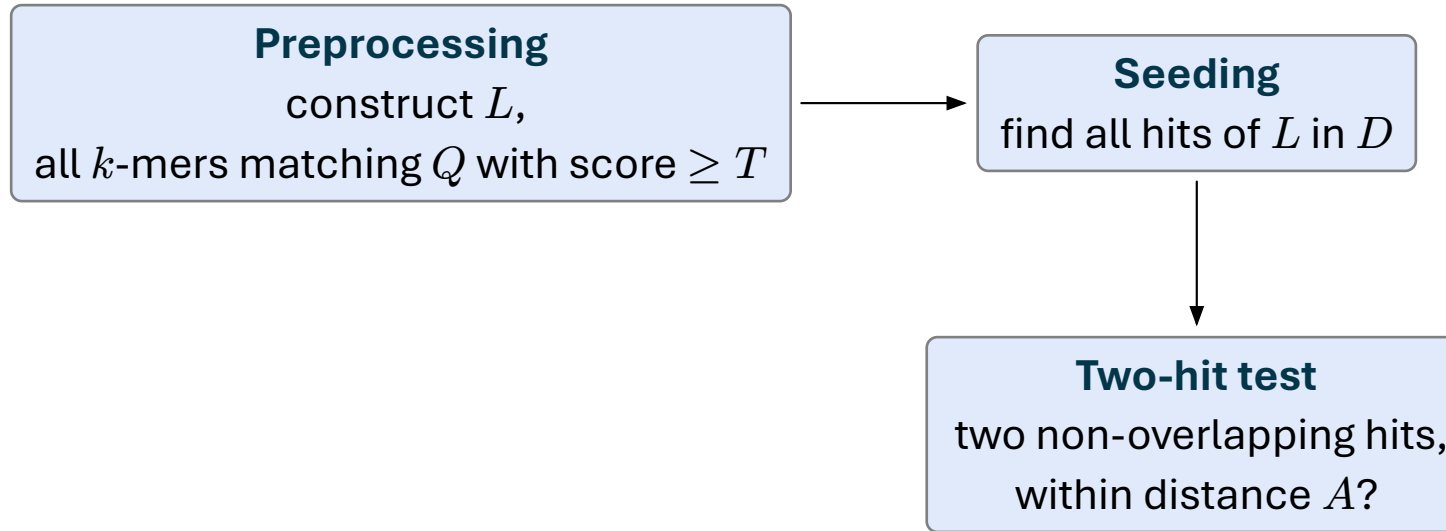
Preprocessing

construct L ,
all k -mers matching Q with score $\geq T$

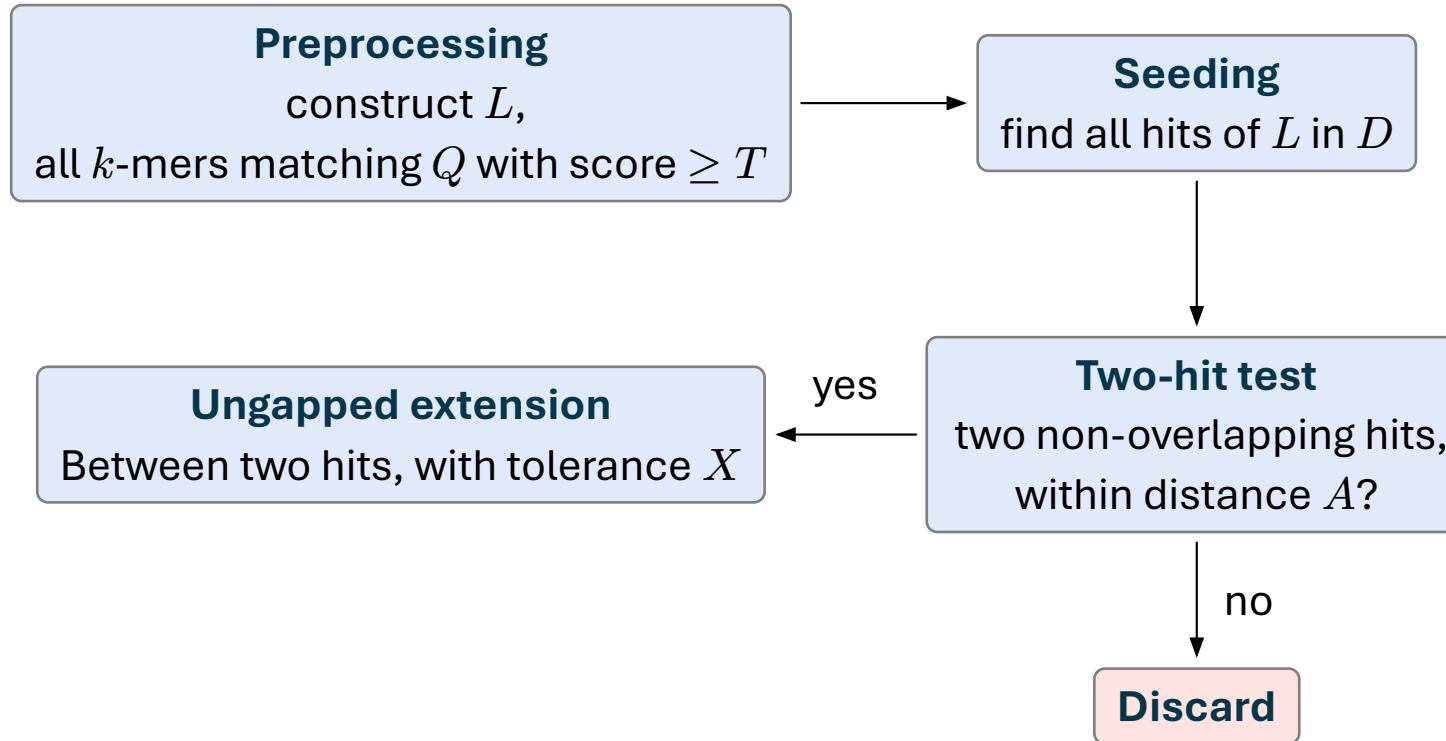
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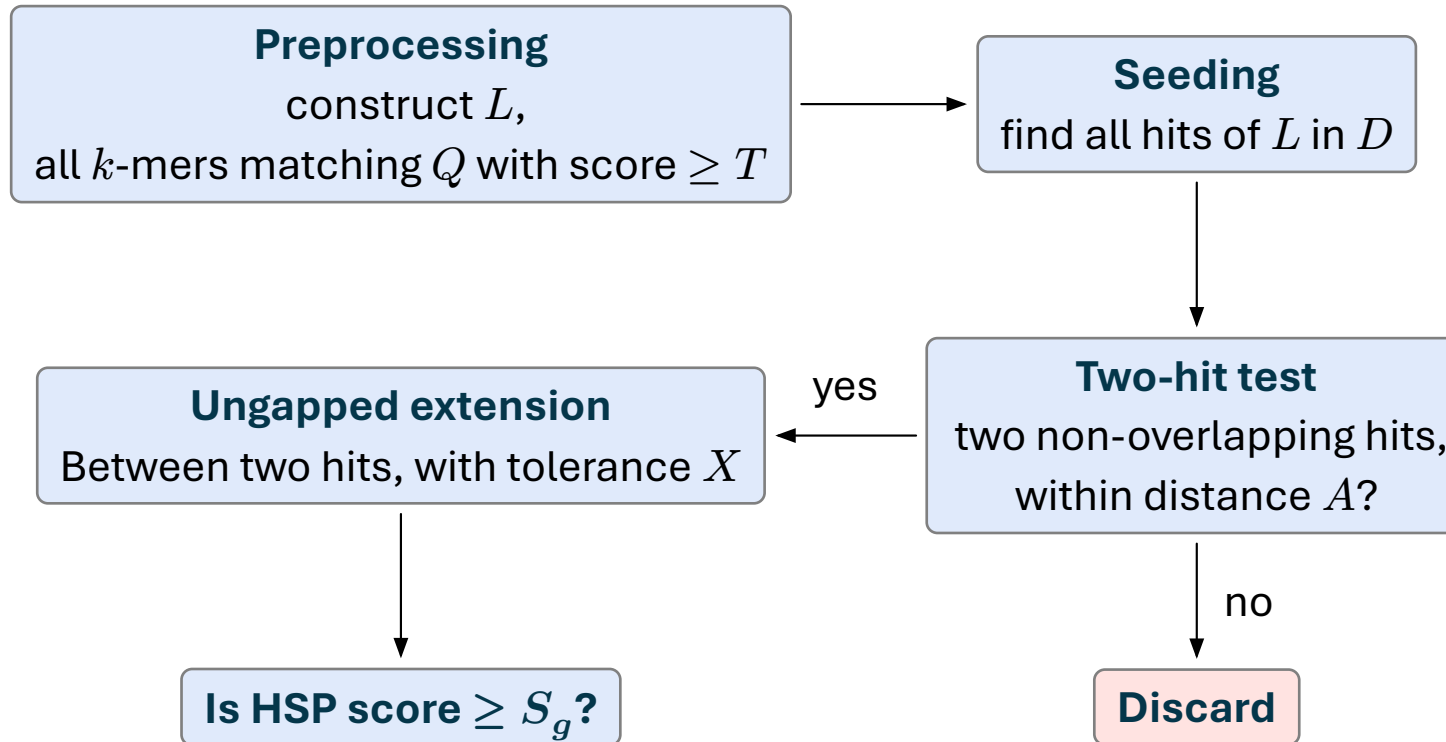
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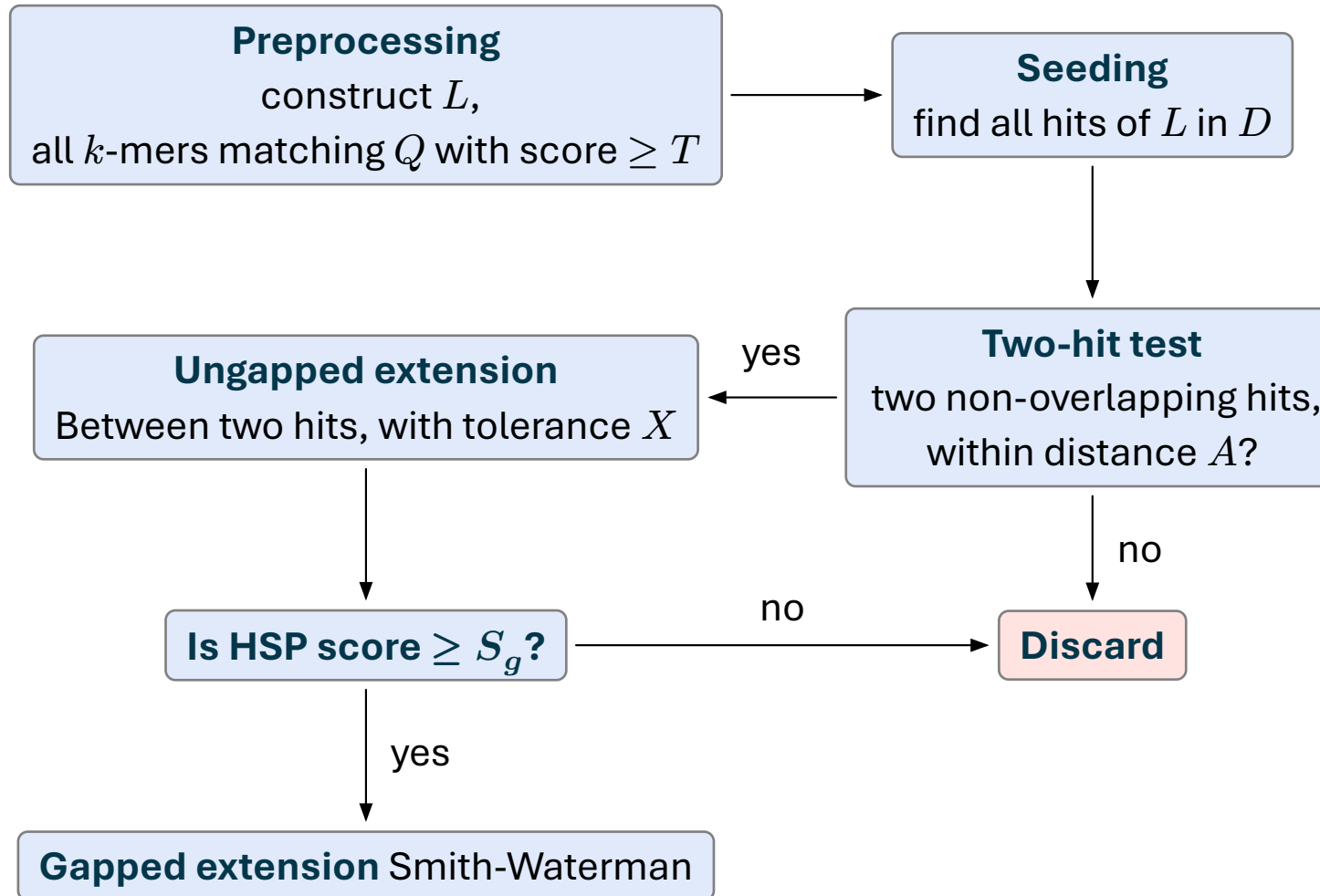
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